

Program : Diploma in Chemical Engineering	
Course Code : 2078	Course Title: CAD Lab
Semester : 2	Credits: 0
Course Category: Engineering Science	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide hands-on experience in Computer Aided Design and Drafting to students using an industry approved CAD software.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Engineering Graphics		Engineering Graphics	1
Computing fundamentals		Introduction to IT systems	1

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize Computer Aided Design.	3	Understanding
CO2	Illustrate the basic commands in AutoCAD	21	Applying
CO3	Sketch block and Isometric 2D drawing in AutoCAD	9	Applying
CO4	Illustrate 3D modelling in AutoCAD	12	Applying

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarize Computer Aided Design.		
M1.01	Introduction to AutoCAD - CAD - Advantages of CAD - History of AutoCAD - Applications of CAD - List commonly used CAD software's in industry - Familiarize AutoCAD drawing editor.	3	Understanding
CO2	Illustrate the basic commands in AutoCAD		
M2.01	Initiate an AutoCAD Drawing Setting the paper size - Space - Setting the limits - Units – Zoom - Use of snap – Grid - Axis - Function keys - Utility commands.	1	Understanding
M2.02	Illustrate Drawing Commands.	2	Understanding
M2.03	Illustrate Text - Hatch – Layers - Line types - Selection of objects.	1	Understanding
M2.04	Illustrate Modify Commands	2	Understanding
M2.05	Illustrate Dimensioning Commands	3	Understanding
M2.06	Sketch the sectional front view and side view of a given object – Sketch minimum four objects.	9	Applying
	Lab Exam – I	3	
CO3	Sketch block and Isometric 2D drawing in AutoCAD		
M3.01	Sketch Block and Title Block.	3	Applying
M3.02	Illustrate Isometric drawing.	3	Applying

M3.03	Sketch the isometric drawing of a given object.	3	Applying
CO4	Illustrate 3D modelling in AutoCAD		
M4.01	Summarize the setting up of 3D drawing in AutoCAD and Viewport settings.	1	Understanding
M4.02	Illustrate 3D Commands in AutoCAD. EXTRUDE – PRESSPULL - REVOLVE - 3D FILLET.	2	Understanding
M4.03	Sketch the 3D drawing of a given object. Sketch minimum 3 objects.	6	Applying
	Lab Exam – II	3	

Text / Reference

T/R	Book Title/Author
T1	Engineering Graphics for diploma; PI Varghese
R1	CAD Lab Manual
R2	Advanced AutoCAD 2013; Cheryl R. Shrock
R3	Engineering Drawing and Graphics + AutoCAD ; K. Venugopal
R4	CAD/CAM/CIM ; P. Radhakrishnan , S. Subramanyan, V. Raju

Online Resources

Sl.No	Website Link
1	https://knowledge.autodesk.com/support/autocad/learn